

Finalise Behaviour-Derived Personas

DA-01-T12 | Feature 1 Handover Document | Behavioural Analytics & Persona Segmentation

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Author Robin Le (An Bich Phuong Le), DA/ML Cross-Team Lead

Approved by Feature 1 team

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Contents

0. Document Control	4
1. Executive Summary	6
2. Feature Purpose & Value Proposition	7
2.1 The problem	7
2.2 Why behavioural segmentation is worth building	7
2.3 The persona-to-recommender value chain	7
2.4 Why this extends, rather than replaces, T1	8
3. Continuity with Trimester 1	9
3.1 What T1 delivered	9
3.2 What T1 recommended for T2	9
3.3 What T2 built on top	9
3.4 The T1 → T2 → T3 arc	10
4. Final Persona Definitions	11
4.1 Cluster 0 — Product-Focused Explorers	11
4.2 Cluster 1 — Deal & Comparison Shoppers	12
4.3 Cluster 2 — Low-Engagement Users	13
4.4 Cluster 3 — Highly Engaged Planners	14
4.5 At-a-glance summary	15
5. Methodology	16
5.1 Feature engineering	16
5.2 Preprocessing	17
5.3 Modelling	17
5.4 Cluster interpretation	18
5.5 Pipeline overview	18
6. Downstream Interface Specification	20
6.1 Output artefact — persona-assignment CSV in the DiscountMate GitHub repo	20
6.2 Proposed interface with AI/DL for DL-05 Personalised Top-N Recommender	21
6.3 Interface with DA-02 Insight Surfacing	23
6.4 Interface with DA-03 Pricing Intelligence	23
6.5 Consumer FAQ	24
7. Assumptions & Limitations	26
7.1 Data volume — synthetic modelling substrate	26
7.2 Feature coverage — Phase 2 GA4 events not yet instrumented	26
7.3 Reporting window — static 28-day rolling, not real-time	26
7.4 No purchase data linked to behavioural signals	26
7.5 Behavioural labels ≠ motivational personas	26
8. Recommendations for Downstream Teams	27
8.1 AI/DL — DL-05 Personalised Top-N Recommender	27
8.2 DA-02 Insight Surfacing	27
8.3 DA-03 Pricing Intelligence	28
8.4 App Dev / Data Engineering — Phase 2 GA4 instrumentation	28

8.5 Incoming T3 Feature 1 lead.....	29
9. Roadmap Forward.....	30
9.1 Near term (T3).....	30
9.2 Medium term	30
9.3 Long term.....	30
10. Appendices.....	32
Appendix A — Data dictionary reference.....	32
Appendix B — Notebook & artefact index	32
Appendix C — Task ledger, DA-01-T1 → T12.....	33
Appendix D — Glossary	34
Appendix E — Change log.....	35

0. Document Control

This section records what this document is, who produced and approved it, who it is for, and which artefacts it draws on. It exists for auditability: a reader should be able to trace every claim in Sections 1–10 back to a named Feature 1 artefact.

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Naming conventions used throughout. “T1”, “T2” and “T3” on their own refer to DiscountMate trimesters (Trimester 1, 2 and 3 of 2026 onwards). Task tickets always carry the full prefix — for example DA-01-T3 is Feature 1, Task 3 — so “T3” alone is never a task. DA-01, DA-02 and DA-03 are the three DA/ML features for T2 2026: Behavioural Analytics & Persona Segmentation (this feature), Insight Surfacing (Power BI dashboards), and Pricing Intelligence (the True Value Promotion model, TVP). DL-05 is the AI/DL team’s Personalised Top-N Recommender. Wn denotes week *n* of the trimester. GA4 is Google Analytics 4, the web analytics platform that records user activity on DiscountMate. Further terms are defined in Appendix D.

Related artefacts

Artefact	Descriptor	Owner
DA-01-T1 GA4 Event Audit Report	Audit of GA4 events live on DiscountMate; identifies the missing behavioural signals later specified as Phase 2 events	Noel Mathew Shillow
DA-01-T2 Technical Review of the Rule-Based Persona Framework	Review of the T1 persona framework; recommends extending rather than replacing it	Carlin Monsy
DA-01-T3 Behavioural Feature Specification v1.0	Defines the behavioural features, preprocessing, clustering approach and T1 continuity hypothesis	Robin Le
DA-01-T4 Behavioural data extraction	Initial behavioural dataset extracted from the GA4 BigQuery export	Carlin Monsy
DA-01-T5 Behavioural data cleaning	Cleaned dataset; invalid records, duplicates and noise removed	Tanvi
DA-01-T6 Normalised clustering-	Min-max-scaled (mm_) feature matrix used at model fit	Shubhpreet Kaur

Artefact	Descriptor	Owner
ready dataset		
DA-01-T7 Behavioural Data Dictionary	Definitions, sources and derivations for each behavioural feature	Noel Mathew Shillow
DA-01-T8 Dataset validation	Quality and schema checks before clustering	Feature 1 team
DA-01-T9 Clustering model notebook (DiscountMate_Clustering_Model.ipynb)	Executable K-means and DBSCAN model	Shubhpreet Kaur
DA-01-T10 Modelling Documentation	Cluster-quality evaluation and model selection (K=4, silhouette ≈ 0.484)	Shubhpreet Kaur
DA-01-T11 Interpret Behavioural Clusters vs T1 Personas	Plain-language cluster interpretation and comparison with the five T1 personas	Robin Le
DA-01 Persona Archetype Definitions and synthetic behavioural dataset	Archetype targets given to DE, and the 1,500-user synthetic dataset DE generated from them for pipeline validation	DA/ML (definitions); DE (dataset)
DA-02-T8 Persona Visualisation Prototype (.pbix)	Power BI prototype visualising the DA-01 clusters	DA-02 (Batnasan)
DiscountMate T2 2026 Roadmap	Source of the DA-01-T1 → T12 task plan, priorities and cross-team dependencies	DA/ML leads
T1 Rule-Based Persona Framework documentation	Adam Creek's T1 exploratory persona document: five personas, questionnaire, static rules	Adam Creek (T1)
DiscountMate Personalised Recommendation Engine, T2 2026 Technical Handover	AI/DL team's DL-05 handover: model contract, 55 features, results, integration plan. Basis for the v1.1 revision of Sections 6.2 and 8.1	AI/DL (Ethan Fu et al.)
DataBytes Project Handover, T1 2026	T1 handover, including the GA4 implementation and Power BI foundation this feature builds on	T1 DataBytes team

File locations for these artefacts are indexed in Appendix B.

1. Executive Summary

Context. DiscountMate helps shoppers compare grocery prices across retailers. To show each shopper the products and deals most relevant to them, the product first needs to know what kinds of shoppers it has. In Trimester 1, the team grouped users into five personas from a sign-up questionnaire and fixed rules. That approach works from a user's first visit, but it never learns from what users actually do on the site. Feature 1 set out to add that missing evidence.

What we built. A repeatable pipeline that takes website activity recorded by Google Analytics 4, turns it into twelve behavioural measures per user (how often they visit, how long they stay, which kinds of pages they favour, how widely they browse), and groups users whose behaviour looks alike.

Result. The pipeline produces four behaviour-based segments:

- **Deal & Comparison Shoppers** (35% of users) — compare products and check specials heavily.
- **Highly Engaged Planners** (25%) — the most active users, with heavy use of shopping lists.
- **Product-Focused Explorers** (20%) — read product pages in depth and browse categories.
- **Low-Engagement Users** (20%) — visit rarely and briefly; a group the T1 personas did not anticipate.

The segments separate with a quality score of 0.484 on a 0-to-1 scale (the silhouette score): moderate, and clear enough for directional use. One segment matches a T1 persona strongly (Deal & Comparison Shoppers with Budget-Conscious Shopper) and one moderately (Highly Engaged Planners with Family Planner). One important caveat: only 28 real users were available, so the model was built and tested on a 1,500-user synthetic dataset. This proves the pipeline works; it does not yet prove these are DiscountMate's real segments.

Value. The segments are an input layer for personalisation. Their main consumer is the AI team's product recommender, which can only rank products a user touched in the last two days. The segments give it a fallback for users it cannot yet score, and extra products to rank. They also feed the stakeholder dashboards and a persona-aware version of the True Value Promotion score, which rates how good a deal really is. The T1 questionnaire personas stay in place as the fallback for new users, so nothing that works today is lost.

What's next (Trimester 3). Three actions carry the most weight:

1. **Re-validate on real data** once roughly 500 active users are recorded over a 28-day window.
2. **Agree the data handover with the AI team.** The format in this document is a proposal, not yet an agreement.
3. **Add the eight missing tracking events.** This is the biggest single lever on segment quality: it would let the segments explain *why* people shop, not only *what* they do.

2. Feature Purpose & Value Proposition

2.1 The problem

DiscountMate's promise is to help shoppers decide where and what to buy. Personalisation is how that promise scales: a shopper who hunts specials, one who plans a weekly list, and one who researches individual products each need a different set of products, deals and surfaces put in front of them. Without a way to tell these users apart, every user gets the same ranking and the product relies on the shopper to do the filtering.

T1 answered this with the Rule-Based Persona Framework (RBPF): five predefined personas, assigned from a self-selection onboarding questionnaire using deterministic rules. As the DA-01-T2 review found, this was the right call for T1. It needs no behavioural history, so it personalises from a user's first session (it solves the cold-start problem), and every recommendation can be traced back to an explicit rule. Its limits are equally clear: persona assignments are static, the rules and keyword dictionaries are maintained by hand, and the framework does not learn from how users actually browse. It relies on what users *say* about themselves rather than what they *do*, and DA-01-T3 cites research showing a persistent gap between the two.

2.2 Why behavioural segmentation is worth building

The product case for Feature 1 rests on three points.

It turns observed behaviour into a signal the product can act on. GA4 already records every session, page view and page type. Feature 1 converts that raw activity into twelve comparable behavioural measures per user and groups users on them. The result is a segment label grounded in evidence, not self-report, and it can be recalculated as behaviour changes.

It tests the T1 personas against evidence. Where a behavioural segment agrees with a T1 persona, that persona is validated. Where it splits or crosses T1 boundaries, the questionnaire may be asking the wrong question. Section 4 shows all three outcomes: one strong alignment, one moderate, one partial, and one segment (Low-Engagement Users) that T1 never defined.

It is cheap to consume and built to improve. The output is one small, version-controlled file keyed on the GA4 user identifier (Section 6.1). Downstream teams can adopt it without taking on the modelling. The schema already reserves space for the Phase 2 features, so segment quality can rise as new tracking events land without re-architecting anything upstream (Sections 7.2 and 8.4).

Stated plainly, the value is conditional. The four segments are directional until they are re-validated on real users (Section 7.1), and they describe behaviour, not motivation (Section 7.5). The case for building now is that a working, validated pipeline is ready the moment real volume arrives. The alternative, waiting for data before building, would leave DiscountMate with nothing to run when the volume comes.

2.3 The persona-to-recommender value chain

Value flows through six links. **GA4 events** (page views, sessions, engagement time, and visits to Compare, Specials and My Lists pages) are exported to BigQuery. The Feature 1 pipeline turns them into **behavioural features**: twelve per-user measures of engagement intensity, page-type affinity, browsing breadth, and recency and tenure, each scaled to a 0–1 range. K-means clustering groups users into **clusters** on those features. Interpretation turns each cluster into a plain-language **persona label** (Section 4). The labels are published in the persona-assignment file, which the **DL-05 Personalised Top-N Recommender** uses in two places. It serves a persona-based Top-N when a user has no recent product interactions, and it widens the pool of candidate products it ranks (Section 6.2). DL-05 then produces the **ranked products the user sees**. The final link is where the value lands: a user sees a relevant list instead of an empty one, or a better-ranked

list instead of a narrow one. DL-05 scores purchase likelihood from the last few days of product interactions, while the persona carries the slower, longer view of how the user shops. Together they cover each other's gaps.

Two further consumers read the same persona-assignment file in parallel. **DA-02 Insight Surfacing** shows the segments in stakeholder Power BI dashboards: segment sizes, defining features and, once refreshes accumulate, movement between segments. **DA-03 Pricing Intelligence** layers per-persona weighting on the core True Value Promotion score to produce a persona-aware variant. For example, it weights discount depth more heavily for Deal & Comparison Shoppers. Because all three consumers read one artefact, a single refresh of the personas updates recommendations, dashboards and value scoring together (Section 6).

2.4 Why this extends, rather than replaces, T1

Behavioural segmentation and the T1 framework do different jobs, and the product needs both. Behavioural segments need activity history: a new user has none, and a Low-Engagement user has too little to personalise on reliably. The T1 questionnaire personas need no history. Keeping the RBPF in the recommender's fallback order, after the behavioural persona and before popularity defaults (Section 6.2), means a new user still gets a personalised first experience, and the product loses nothing it has today.

Explainability is kept in the same way. Each behavioural segment is described in plain language by its defining features, and each is mapped back to the T1 personas with an explicit alignment grade: Strong, Moderate, Partial or None (Section 4). A product owner can still say why a user sees what they see.

What the behavioural layer adds is evidence. It confirms the T1 persona that behaviour supports most directly (Budget-Conscious Shopper), qualifies the ones it cannot yet confirm, and surfaces a segment the rules never anticipated. This is the direction DA-01-T2 recommended and DA-01-T11 confirmed: use behavioural evidence to validate and progressively refine the persona framework, not to discard it.

3. Continuity with Trimester 1

3.1 What T1 delivered

Adam Creek's Rule-Based Persona Framework is the foundation this feature builds on. It defined five personas: Budget-Conscious Shopper, Health Enthusiast, Family Planner, Convenience Seeker and Premium Shopper. Users were assigned through self-selection or an onboarding questionnaire evaluated by deterministic decision rules. Products were linked to personas through manually curated keyword dictionaries, and a product could belong to more than one persona. The T1 documentation itself described the RBPF as an initial implementation and pointed towards behavioural data and hybrid recommendation as future work.

Two other T1 foundations made Feature 1 possible. Adam Creek implemented GA4 across the DiscountMate web application in T1, which produced the event stream that Feature 1 now models. Steve Nguyen built the Power BI semantic layer and deployed the customer-facing dashboard, the base that DA-02 extends in T2 and through which the behavioural segments are visualised.

3.2 What T1 recommended for T2

Carlin Monsy's DA-01-T2 technical review set the direction for T2. It confirmed the RBPF's strengths (transparent logic, no need for historical data, rapid implementation) and named its limits: static persona assignment, hand-maintained rules and keyword dictionaries, limited behavioural learning, and scaling pressure as the catalogue grows. Its recommendations to DA-01-T3 were to:

- retain the RBPF as the baseline approach
- validate the predefined personas against real behaviour as data becomes available
- refine keyword mappings as the catalogue expands
- use the analytics infrastructure T1 had built
- keep the framework's explainability while introducing more adaptive techniques step by step

In short: **extend, don't replace.**

3.3 What T2 built on top

T2 added a behavioural evidence layer on top of the T1 framework and did not rebuild it. The T2 Roadmap framed Feature 1 as a shift from the rule-based framework to GA4-derived clustering. The DA-01-T2 review and the DA-01-T11 findings refined that framing into an extension, and this handover adopts it.

Three things were added:

- **Behavioural evidence.** DA-01-T1 audited the live GA4 events, and DA-01-T3 specified twelve Phase 1 behavioural features that could be built from them, plus the Phase 2 features that need new events (Section 5.1).
- **Model-based clustering.** The data passed through extraction, cleaning, normalisation and validation. K-means was then fitted and selected at K=4. DBSCAN (Density-Based Spatial Clustering of Applications with Noise) was tested as an alternative and documented. PCA (Principal Component Analysis) served as a visual check (Section 5).
- **Mapping back to T1.** DA-01-T3 wrote the expected behavioural signature of each T1 persona as a hypothesis. DA-01-T11 then tested the actual clusters against it as a comparison, without forcing any cluster onto a T1 label (Section 5.4). The result, in Section 4, keeps the T1 personas as context for each behavioural label.

3.4 The T1 → T2 → T3 arc

The feature follows a three-stage arc. In **T1**, DiscountMate personalised with rules: five personas, assigned by questionnaire, transparent and usable from the first session, but static. In **T2**, this feature added a behavioural evidence layer: a working pipeline from GA4 events to four interpretable behavioural segments, each mapped back to the T1 personas, with the RBPF kept as the cold-start fallback. The T2 layer is validated on synthetic data and is ready for real data. In **T3**, the task is integration and proof. That means agreeing the interface with AI/DL, using the segments in the DL-05 recommender, re-validating them on real GA4 users, and adding the Phase 2 events that let segments speak to motivation (Sections 8 and 9). Beyond T3, the long-term direction inherited from the T1 handover is a hybrid persona system that combines rule-based, behavioural and purchase-driven signals (Section 9.3).

4. Final Persona Definitions

This section presents the four behaviour-derived personas ratified by the Feature 1 team for T2 2026. They are the final outputs of DA-01-T10 (K-means, K=4, silhouette $\approx$ 0.484 on a 1,500-user market-research-grounded synthetic dataset) and DA-01-T11 (cluster interpretation and T1 comparison), and they supersede any earlier working labels used during model development.

The persona names are deliberately behaviour-based (Product-Focused Explorers, Deal & Comparison Shoppers, Low-Engagement Users, Highly Engaged Planners) rather than persona-style in the T1 mould (Family Planner, Health Enthusiast, etc.). The current feature set covers engagement intensity, page-type affinity, and browsing breadth only; it does not yet cover category, retailer, or price-alert behaviour. Behaviour-based labels state what the data supports and no more. Richer persona-style labels — the kind that speak to underlying motivation — require the Phase 2 GA4 events specified in DA-01-T3 (Product_Click, Retailer_Selected, Filter_Used, Price_Alert_Created, Add_To_Favourites, Coupon_Click) and further behavioural research; these are proposed as T3 uplift work.

Each persona card that follows carries: cluster ID, label, one-line description, size, defining behavioural signature (top mm_ features with direction, sourced from DA-01-T10 cluster profiles and interpreted in DA-01-T11), mapping to the T1 Rule-Based Persona Framework (per DA-01-T11), recommender implication for the DL-05 personalised Top-N recommender (reasoned from the behavioural signature; the downstream contract is proposed in Section 6 and is pending AI/DL confirmation), and a confidence note.

Reading the signature: mm_ feature values are min-max scaled to [0, 1] across the 1,500-user modelling matrix, so 0.75 means the persona sits at the 75th percentile of that feature across all users, not that 75% of the persona does the thing. H = high (well above the population mid-point), L = low (well below), M = moderate.

4.1 Cluster 0 — Product-Focused Explorers

Cluster ID	0
Persona label	Product-Focused Explorers
One-line description	Users who spend time reading product detail pages and browsing categories, with high per-session engagement but limited deal- or list-driven behaviour.
Size	300 users (20.0% of 1,500)
Source	DA-01-T10 cluster profiles; DA-01-T11 §3.1, §5.1

Defining behavioural signature

Feature (mm_ scaled)	Value	Direction	What it says
mm_product_detail_ratio	0.76	H	High share of pageviews are product-detail pages
mm_avg_engagement_time_sec	0.70	H	Long per-session engagement time
mm_category_browse_ratio	0.68	H	High share of category browse pages
mm_browsing_entropy	0.56	M	Moderate spread across page types
mm_compare_ratio	low	L	Little comparison-page activity
mm_specials_ratio	low	L	Little specials/deals activity

Mapping to T1 personas (per DA-01-T11 §5.1)

T1 Persona	Alignment	Note
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T1 Persona	Alignment	Note
Health Enthusiast	Partial	Product-detail focus is compatible with health-oriented product research, but current features cannot establish health intent.
Convenience Seeker	Partial	Sustained engagement could also indicate convenience-led evaluation; again, not distinguishable with current features.
Budget-Conscious Shopper	None	Low compare/specials activity rules this out.
Family Planner	None	No list-building signal.
Premium Shopper	None	No premium-category signal available.

Recommender implication (proposed for DL-05; candidate and fallback rules per Section 6.2, pending AI/DL confirmation)

Persona candidate rule (integration point 2 in Section 6.2): products similar to, or in the same categories as, those the user has recently viewed in detail, plus products popular in those categories. Put less weight on deal-first candidates. Category-affinity candidates are the highest-value addition for this segment once Phase 2 Product_Click and Add_To_Favourites events provide product-level signal. These events also supply the product IDs DL-05 needs to score these users with its own model.

Confidence note

Cluster is well-separated in PCA space (per DA-01-T10, first two components explain ≈67.9% of variance) and interpretable. Partial-only mapping to two T1 personas is a known feature-set limitation, not a modelling weakness. Confidence in the behavioural signature is moderate-to-high; confidence in any motivational label (health vs convenience) is low and should not be asserted downstream until Phase 2 events land.

4.2 Cluster 1 — Deal & Comparison Shoppers

Cluster ID	1
Persona label	Deal & Comparison Shoppers
One-line description	The largest segment. Users who actively compare products and check specials, with the widest browsing spread across page types.
Size	525 users (35.0% of 1,500)
Source	DA-01-T10 cluster profiles; DA-01-T11 §3.2, §5.2

Defining behavioural signature

Feature (mm_ scaled)	Value	Direction	What it says
mm_browsing_entropy	0.85	H	Very varied browsing across page types
mm_compare_ratio	0.73	H	High share of pageviews are compare pages
mm_specials_ratio	0.72	H	High share of pageviews are specials pages
mm_unique_pages_visited	0.54	M	Moderately broad set of unique pages
mm_mylists_ratio	low	L	Little list-building activity
mm_product_detail_ratio	low	L	Less time on any single product page

Mapping to T1 personas (per DA-01-T11 §5.2)

T1 Persona	Alignment	Note
Budget-Conscious Shopper	Strong	High compare and specials activity is direct behavioural evidence of

T1 Persona	Alignment	Note
		value-oriented, deal-seeking shopping — the clearest cluster-to-T1 alignment in the study.
Family Planner	None	No list-building signal.
Health Enthusiast	None	No category-affinity signal.
Convenience Seeker	None	Broad browsing contradicts convenience-led shortcut behaviour.
Premium Shopper	None	Deal-seeking behaviour is inconsistent with premium orientation.

Recommender implication (proposed for DL-05; candidate and fallback rules per Section 6.2, pending AI/DL confirmation)

Persona candidate rule (integration point 2 in Section 6.2): current specials, cross-retailer price gaps, and items with a high DA-03 True Value Promotion V2 score (see 6.4). These candidates can be built from the product and pricing database today, before product-level GA4 exists. That makes this segment the natural first test of the persona fallback and candidate source.

Confidence note

Highest-confidence cluster in the study: largest sample (n=525), strongest T1 alignment, and the behavioural signature is directly relevant to DiscountMate's core value proposition. Safe to prioritise for the first DL-05 iteration.

4.3 Cluster 2 — Low-Engagement Users

Cluster ID	2
Persona label	Low-Engagement Users
One-line description	Users with limited activity across most behavioural features and no dominant interaction pattern. Defined by low engagement, not by a shopping preference.
Size	300 users (20.0% of 1,500)
Source	DA-01-T10 cluster profiles; DA-01-T11 §3.3, §5.3

Defining behavioural signature

Feature (mm_ scaled)	Value	Direction	What it says
mm_browsing_entropy	0.15	L	Narrow browsing — few page types touched
mm_category_browse_ratio	0.13	L	Very little category browsing
mm_product_detail_ratio	0.09	L	Very little product-detail viewing
mm_sessions_count	low	L	Few sessions overall
mm_avg_engagement_time_sec	low	L	Short session engagement
mm_days_since_last_activity	0.29	M	Longer gap since last visit than active segments

Mapping to T1 personas (per DA-01-T11 §5.3)

T1 Persona	Alignment	Note
(No T1 persona)	None	The defining trait is limited engagement rather than a shopping preference. T1 did not define an engagement-based segment; this is a genuinely new segment surfaced by behavioural clustering.

Recommender implication (proposed for DL-05; candidate and fallback rules per Section 6.2, pending AI/DL confirmation)

Insufficient behavioural signal for personalised ranking, and the segment least likely to have any recent-interaction candidates for DL-05. The recommended default is a popularity- or trending-based Top-N (site-wide, or by retailer if location is available), or current specials. This is step 4 of the fallback order in 6.2, applied even when a persona label exists. If the user answered the T1 onboarding questionnaire, the T1 RBPF Top-N (step 3) is an acceptable alternative.

Confidence note

Cluster is stable in the current data but low-engagement segments are inherently harder to act on. Product-level implication: this segment is a candidate for reactivation campaigns and onboarding UX work, not for aggressive personalisation. On real GA4 data, some users in this cluster may prove to be lightly-instrumented rather than genuinely low-engagement — worth re-checking once Phase 2 events increase per-user signal.

4.4 Cluster 3 — Highly Engaged Planners

Cluster ID	3
Persona label	Highly Engaged Planners
One-line description	Users with the highest overall activity, strong My Lists usage, many sessions, and broad page coverage. An organised, repeated-use pattern.
Size	375 users (25.0% of 1,500)
Source	DA-01-T10 cluster profiles; DA-01-T11 §3.4, §5.4

Defining behavioural signature

Feature (mm_ scaled)	Value	Direction	What it says
mm_mylists_ratio	0.74	H	Distinctively high share of pageviews are MyLists
mm_browsing_entropy	0.74	H	Broad spread across page types
mm_unique_pages_visited	0.72	H	Many distinct pages per user
mm_sessions_count	0.71	H	High session count over the reporting window
mm_pageviews_per_session	0.68	H	High pageviews within each session

Mapping to T1 personas (per DA-01-T11 §5.4)

T1 Persona	Alignment	Note
Family Planner	Moderate	Strong MyLists use and repeated broad engagement are compatible with organised planning behaviour, but the current features do not establish family status.
Budget-Conscious Shopper	None	Not defined by compare/specials — a different flavour of high engagement.
Health Enthusiast / Convenience Seeker / Premium Shopper	None	No category-affinity signal to support these.

Recommender implication (proposed for DL-05; candidate and fallback rules per Section 6.2, pending AI/DL confirmation)

Persona candidate rule (integration point 2 in Section 6.2): items already on the user's lists, staples and repeat-purchase candidates. This maps to the “previously purchased products due for repeat purchase”

source in the AI/DL handover, with list items standing in for purchases on DiscountMate. This segment is likely to have the most recent-interaction candidates, so DL-05's own ranking should carry most of the load here, and the persona adds candidates rather than acting as a fallback. It is highly compatible with the DA-03 persona-aware TVP variant (see 6.4) if planning users also value the True Value signal on their staples.

Confidence note

High confidence in the behavioural signature and in the recommender implication. Moderate confidence in the Family Planner mapping — the interaction pattern is consistent with it but does not prove it. Downstream teams should treat "Highly Engaged Planners" as the primary label; the T1 mapping is context, not identity.

4.5 At-a-glance summary

Cluster	Persona label	N	% of 1,500	Dominant signal	Closest T1	T1 alignment
0	Product-Focused Explorers	300	20.0%	Product detail + category browse	Health Enth. / Conv. Seeker	Partial
1	Deal & Comparison Shoppers	525	35.0%	Compare + Specials + broad browsing	Budget-Conscious Shopper	Strong
2	Low-Engagement Users	300	20.0%	Low activity across most features	(none)	None
3	Highly Engaged Planners	375	25.0%	MyLists + sessions + pages + entropy	Family Planner	Moderate

Note: Premium Shopper (T1) has no clear behavioural equivalent in the current K=4 solution — no cluster is primarily characterised by premium-product preference. This is documented in DA-01-T11 §5.5 and is a known limitation of the Phase 1 feature set; it is not a data-quality issue.

5. Methodology

This section documents the modelling pipeline that produced the four personas in Section 4. It is a summary suitable for downstream reference; the executable artefacts are the DA-01-T10 clustering notebook (DiscountMate_Clustering_Model.ipynb) and Modelling_Documentation.pdf, and the definitional artefact is DA-01-T3 (Behavioural Feature Specification). Where a detail is captured elsewhere, this section links to it rather than reproducing it.

Two framings run through every subsection and are worth stating upfront. First, the pipeline is validated against a 1,500-user market-research-grounded synthetic dataset, not against real GA4 behaviour at volume; the original GA4 sample was 28 users per DA-01-T11 §8, well below the ~50–100 floor recommended in DA-01-T3 §6. Second, the current feature set is Phase 1 only — engagement, page-type affinity, and browsing breadth. The Phase 2 events specified in DA-01-T3 (Product_Click, Add_To_Favourites, Price_Alert_Created, Retailer_Selected, Coupon_Click, Filter_Used, Sort_Used, Outbound_Merchant_Click) are not yet instrumented. Both framings shape what the model can and cannot claim, and both are re-stated in Section 7 with their downstream impact.

5.1 Feature engineering

Twelve behavioural features enter the clustering model, each prefixed mm_ to indicate min-max scaling to [0, 1] (see 5.2). The features are grouped by the six behavioural dimensions defined in DA-01-T3 §3, so this handover ties directly back to the spec Shubhpreet, Tanvi, and Carlin worked to.

Dimension (DA-01-T3)	Feature	Phase	What it captures
A — Engagement intensity	mm_sessions_count	1	Session count over the reporting window.
A — Engagement intensity	mm_pageviews_per_session	1	Depth of activity within each session.
A — Engagement intensity	mm_avg_engagement_time_sec	1	Per-session engagement time from GA4 user_engagement.
B — Feature affinity (page-type ratios)	mm_product_detail_ratio	1	Share of pageviews on product-detail pages.
B — Feature affinity (page-type ratios)	mm_category_browse_ratio	1	Share of pageviews on category-browse pages.
B — Feature affinity (page-type ratios)	mm_compare_ratio	1	Share of pageviews on Compare pages.
B — Feature affinity (page-type ratios)	mm_specials_ratio	1	Share of pageviews on Specials pages (discount-interest proxy).
B — Feature affinity (page-type ratios)	mm_mylists_ratio	1	Share of pageviews on MyLists pages.
C — Discovery behaviour	mm_browsing_entropy	1	Shannon-style spread across page types; captures narrow vs. broad browsing.
C — Discovery behaviour	mm_unique_pages_visited	1	Breadth of distinct pages seen per user.
F — Recency & tenure	mm_days_since_first_visit	1	Tenure proxy.
F — Recency & tenure	mm_days_since_last_activity	1	Recency proxy.

Two design decisions are worth calling out for downstream teams reading this cold:

- Ratios over raw counts (Dimension B). Compare_ratio, specials_ratio, mylists_ratio, product_detail_ratio, and category_browse_ratio are all page-type shares of a user's total pageviews, per DA-01-T3 §3.2. Raw counts co-move with overall activity and let heavy users dominate every affinity signal; ratios make a light user with a strong compare bias and a heavy user with a strong compare bias comparable in the same feature space. This is why Deal & Comparison Shoppers separate cleanly from Highly Engaged Planners rather than collapsing into a single "active users" cluster.
- Phase 1 vs. Phase 2. All 12 features above are Phase 1 — they were buildable from GA4 events already live at the time of the T1 audit (page_view, session_start, user_engagement, and the named Compare_Page/Specials_Page/MyLists_Page events). The eight Phase 2 features listed in DA-01-T3 §3.3–3.5 (filter/sort usage, price-alert and coupon signals, retailer diversity and top-retailer share, category-affinity vector, favourites count) are held in the schema with nulls; they will populate the moment App Dev instruments the eight missing events. Section 7 records the persona-quality impact of that gap; Section 8.4 makes the ask.

One deliberate exclusion is worth documenting so future contributors do not re-add it: avg_percent_scrolled is not in the feature set. GA4's default scroll event fires once at 90% depth per page and is not a continuous measurement; treating it as one would misrepresent reading depth. The exclusion is recorded in the DA-01 Persona Archetype Definitions and the synthetic dataset method note; the DA-01-T7 data dictionary still lists the feature among its 13 and should carry the same note (see Appendix A).

5.2 Preprocessing

Preprocessing follows the requirements in DA-01-T3 §4 and is executed in DA-01-T5 (cleaning) and DA-01-T6 (normalisation).

- Aggregation. One row per user_pseudo_id (GA4 client identifier), rolling 28-day reporting window (see 7.3).
- Inclusion. Users with total_pageviews_28d = 0 are dropped (nothing to cluster on). Count-based features impute 0 when the underlying event did not fire; ratio-based features impute 0 when the numerator is 0.
- Normalisation. Min-max scaling to [0, 1] across all 12 features, producing the mm_ columns used at model fit and in every cluster profile. Min-max was chosen over z-score because the DBSCAN comparison run needs an epsilon (ϵ) neighbourhood that is interpretable across features, and because the persona-signature tables in Section 4 read more naturally as "0.76 = 76th percentile" than as z-scores. The unscaled dataset is retained for interpretation in DA-01-T11 and downstream QA.
- Outlier handling. Continuous features are capped at the 99th percentile per feature before scaling to reduce K-means distortion from long tails; DBSCAN handles residual outliers natively as noise points.

The resulting clustering-ready dataset carries user_pseudo_id, reporting_window_start/end, the 12 mm_Phase-1 features, and an is_included_in_run flag, matching the schema in DA-01-T3 §5.1. On the synthetic run, 1,500 users pass inclusion; all are used at fit.

5.3 Modelling

Two algorithms were run in parallel, as DA-01-T3 §6 anticipated, so that the team could report which one carries the segmentation and why.

K-means was the primary method. Different values of K were fit on the 12-feature scaled matrix, and K was selected using the elbow method together with silhouette score. K=4 was selected on the criterion that

silhouette peaked there (≈ 0.484) and the elbow flattened around the same point. The resulting cluster sizes were 300, 525, 300, and 375 — no runt clusters, no dominant single group — which is consistent with a real solution rather than a single-mode dataset carved into arbitrary slices. Silhouette ≈ 0.484 is a moderate score, not a strong one; it is enough to support the four labels as directional segments and is the ceiling one should expect from a 12-feature Phase-1 space. It is documented as-is in Section 4 rather than framed higher.

DBSCAN was run as an alternative and is documented for completeness. With the chosen ϵ/minPts settings it produced one main cluster of 293 users and classified 1,207 users as noise (points/label = -1). That is not a usable segmentation on this dataset — the assignment table would send the majority of users to a "no persona" bucket — so K=4 K-means was retained as the operational model. The negative result is itself informative and belongs in the handover: on a min-max scaled behavioural space of this shape, density-based clustering finds one dense core and a long tail, which is what you would expect if activity is broadly continuous rather than sharply grouped.

PCA was used as a visual check on the K-means solution rather than as a modelling step. The first two principal components explained approximately 67.9% of the variance in the scaled matrix; on the PC1×PC2 projection the four clusters appear as distinct groups. Two components at $\sim 67.9\%$ is a good check without being definitive — it means the clusters are clearly separable in the dominant subspace, not that they exhaust the structure of the data. A heatmap of per-cluster feature means was produced in the same notebook and drove the per-cluster interpretation in DA-01-T11.

Reproducibility. Random seeds are fixed in the notebook; hyperparameters and cluster-assignment table are saved alongside the model artefact (per DA-01-T3 §6). The notebook does not currently write export files (no `to_csv/to_parquet/joblib.dump` calls at time of handover); the persona-assignment CSV described in Section 6.1 will be produced by an export cell added by Shubhpreet before T3.

5.4 Cluster interpretation

Interpretation is DA-01-T11 in full. In summary, per-cluster feature means and the PC1×PC2 heatmap were used to translate each K-means group from a centroid into plain language — "high compare and specials activity with varied browsing", say — and each cluster was compared against the five T1 Rule-Based Persona Framework personas (Budget-Conscious Shopper, Health Enthusiast, Family Planner, Convenience Seeker, Premium Shopper).

The framing throughout DA-01-T11 is comparison, not mapping. Behavioural clusters are compared against T1 personas to confirm, split, or cross them; they are not pre-assigned to T1 labels. All three outcomes are informative and none is a failure state, as DA-01-T3 §7 anticipated. The results, carried forward as Section 4 of this handover, are: Deal & Comparison Shoppers align strongly with Budget-Conscious Shopper; Highly Engaged Planners align moderately with Family Planner; Product-Focused Explorers overlap partially with Health Enthusiast and Convenience Seeker but current features cannot distinguish which; Low-Engagement Users have no T1 equivalent and are a genuinely new engagement-based segment; and no cluster reproduces Premium Shopper. The Premium Shopper absence is a Phase-1 feature-set limitation, not a modelling weakness, and is documented as such — Section 7.2 carries the impact and mitigation.

5.5 Pipeline overview

End to end, the Feature 1 pipeline runs GA4 → BigQuery → extraction (DA-01-T4) → cleaning (DA-01-T5) → normalisation (DA-01-T6) → validation (DA-01-T8) → clustering (DA-01-T9) → evaluation (DA-01-T10) → interpretation (DA-01-T11) → finalisation (DA-01-T12, this handover). The dataset owner shifts across the pipeline — Noel owns the DA-01-T1 event audit and DA-01-T7 data dictionary; Carlin owns DA-01-T4 extraction; Tanvi owns DA-01-T5 cleaning and derived features; Shubhpreet owns DA-01-T6 normalisation

and DA-01-T9–T10 modelling; Robin owns DA-01-T11 interpretation and DA-01-T12 finalisation — and each handover is a named artefact rather than an implicit handshake.

Two upstream inputs and one downstream output define the pipeline’s external contract. Upstream: the GA4 property (which produces raw events into BigQuery via the standard export) and the T1 Rule-Based Persona Framework (which provides the interpretive baseline for DA-01-T11 comparison). Downstream: the persona-assignment CSV specified in Section 6.1 — a flat, diffable, version-controlled artefact in the DiscountMate GitHub repo rather than a live table or an API, deliberately kept small; DL-05 needs only the persona label at first, per the revised interface in Section 6.2. Reference artefacts are `DiscountMate_Clustering_Model.ipynb` (the executable model) and `Modelling_Documentation.pdf` (the model evaluation write-up); this Section 5 does not reproduce their code or numbers.

6. Downstream Interface Specification

STATUS — PROPOSAL, NOT AGREEMENT. This interface specification has NOT been agreed with the AI/DL team (Ethan Fu, Sunain Mushtaq, Lam Yu) or with the downstream consumers of DA-02 and DA-03. T2 time constraints prevented the cross-team interface-contract negotiation from concluding within this trimester. Section 6 is therefore drafted as a proposal from the DA/ML side, tabled for AI/DL and downstream team review in T3. Language throughout is deliberately "proposed", "for review", and "pending confirmation".

This section covers: (6.1) the persona-assignment output artefact; (6.2) proposed interface with AI/DL for DL-05 Personalised Top-N Recommender; (6.3) interface with DA-02 Insight Surfacing; (6.4) interface with DA-03 Pricing Intelligence for the persona-aware TVP scoring variant; and (6.5) a consumer FAQ covering the operational questions a downstream engineer or analyst is most likely to raise.

6.1 Output artefact — persona-assignment CSV in the DiscountMate GitHub repo

The persona-assignment artefact is a flat CSV file checked into the DiscountMate GitHub repository under DA/ML's data area. This is deliberately not a BigQuery table and not a live API in T2 — the intent is to give downstream teams a stable, diffable, version-controlled reference they can pin, and to keep the DA/ML surface area small. On the v1.1 proposal, DL-05 needs only persona_label at first, and the mm_ columns serve the optional ranking-feature test and DA-02/DA-03 (see 6.2).

Proposed location and filename

Repository	DataBytes-Organisation/DiscountMate_new, the same repository as the DL-05 implementation (ML/Recommendation_system/Recommendation-2026-t2/). Proposed; exact DA/ML path to agree with AI/DL in T3
Directory	data/behavioural/personas/
Filename convention	persona_assignments_YYYYMMDD.csv (YYYYMMDD = assignment_date, UTC)
Latest-pointer	personas_latest.csv — symlink or committed copy of the most recent dated file, for consumers that want a stable path (proposed; confirm handling in T3).
Encoding / format	UTF-8, RFC 4180 CSV, header row on line 1, LF line endings, no BOM.
Reference notebook	notebooks/DiscountMate_Clustering_Model.ipynb (DA-01-T10 modelling artefact)

Proposed columns

Column	Type	Notes
user_pseudo_id	string	GA4 client identifier; primary key on the file
cluster_id	int (0–3)	As produced by DA-01-T10 K-means model
persona_label	string	Human-readable label from §4 (e.g. "Deal & Comparison Shoppers")
assignment_date	date (YYYY-MM-DD)	Date the notebook run produced this assignment
reporting_window_start	date	Start of the 28-day GA4 window used for features
reporting_window_end	date	End of the 28-day window
mm_sessions_count	float [0,1]	Min-max scaled; 12 mm_ features below all inclusive
mm_avg_engagement_time_sec	float [0,1]	

Column	Type	Notes
mm_pageviews_per_session	float [0,1]	
mm_compare_ratio	float [0,1]	
mm_specials_ratio	float [0,1]	
mm_mylists_ratio	float [0,1]	
mm_category_browse_ratio	float [0,1]	
mm_product_detail_ratio	float [0,1]	
mm_unique_pages_visited	float [0,1]	
mm_browsing_entropy	float [0,1]	
mm_days_since_first_visit	float [0,1]	
mm_days_since_last_activity	float [0,1]	

Refresh cadence

Current (T2 close)	Manual re-run of DiscountMate_Clustering_Model.ipynb by DA/ML on request. No automation.
Proposed T3 target	Weekly refresh, aligned with the GA4 BigQuery daily export cadence and the DE silver-layer refresh schedule. Trigger: scheduled job (GitHub Action or GCP Cloud Scheduler, for DE/DA to confirm in T3). Assignment_date is the run date.
Interim fallback	If T3 automation slips, DA/ML commits to a fortnightly manual refresh and publishes the refresh date in the file's git commit message.

6.2 Proposed interface with AI/DL for DL-05 Personalised Top-N Recommender

Revised in v1.1 after reviewing the AI/DL team's T2 2026 technical handover for the Personalised Recommendation Engine (Ethan Fu et al.; merged as PR #319). This is still a DA/ML proposal. AI/DL has not agreed to any part of it.

What DL-05 is at T2 close (per the AI/DL handover)

DL-05 is an offline pipeline that ranks products for a user by predicting purchase. For every candidate user–product pair, a calibrated Logistic Regression model estimates the probability that the user buys that product in the next three days. It then sorts the candidates and returns a Top-10 list. The model reads a fixed, ordered set of 55 features: user–product behaviour, the user's profile, product, category and brand popularity, recency, and price. It was trained on REES46, a public e-commerce dataset (October 2019, 42.4 million events), not on DiscountMate data.

On its held-out test period, ranking is strong for the products it actually considers: 96.9% of products bought later were ranked in the Top-10 when they were in the candidate set. The weak point is which products it considers. Candidates are limited to products the user interacted with in the last two days. As a result, only 11.7% of candidate pairs and 13.4% of users are covered, and overall HitRate@10 is 13.1%. The pipeline is not yet connected to live GA4 data, because current GA4 events carry no product IDs. REES46 product IDs also do not map to DiscountMate's product database, and no serving API has been built. When a user has no candidates, the draft serving code returns an empty list, and the handover flags a fallback source as future work. The AI/DL handover's own priority list includes *broader candidate generation* (Priority 6) and *cold-start fallbacks, including "broad personas/segments"* (Priority 7).

What this changes for DA/ML's proposal

The v1.0 proposal assumed DL-05 would take `cluster_id` and `persona_label` as a feature that conditions its ranking. The model's 55-feature contract rules this out for now: the feature list is fixed and ordered, and adding a persona input means retraining. The personas are therefore most useful at the two points the AI/DL handover itself names as gaps. These are the empty-candidate fallback and additional candidate sources. They are also useful as a lens for evaluating DL-05, and later as an optional ranking feature once DL-05 is retrained on DiscountMate data.

Integration point	What DA/ML provides	What DL-05 does with it	When
1. Fallback when a user has no candidates (AI/DL Priority 7)	<code>persona_label</code> per <code>user_pseudo_id</code> (Section 6.1)	For users with no recent product interactions, serves a persona-based Top-N instead of an empty list (fallback order below)	T3, first. Can run before product-level GA4 exists
2. Extra candidate source (AI/DL Priority 6)	Per-persona candidate rules from the Section 4 cards (e.g. specials and high-TVP items for Deal & Comparison Shoppers)	Adds persona candidates to the recent-interaction pool, then ranks them with the existing model where features can be computed	T3, once product-level GA4 events flow
3. Evaluation by segment	<code>persona_label</code> joined to test users	Reports NDCG@10, HitRate@10 and candidate coverage for each persona, not only as a global figure	Any offline evaluation run
4. Ranking feature (optional)	<code>cluster</code> flags and/or the 12 <code>mm_</code> features on the same row	Tests the 55 features plus persona inputs against the 55 alone when retraining on DiscountMate behaviour	After the DiscountMate retrain (AI/DL Priority 5)

Integration points 1 and 2 use only `persona_label`. The `mm_` columns stay on the file for integration point 4 and for DA-02 and DA-03.

Proposed join and data contract

- **Join key.** DA-01 keys on GA4 `user_pseudo_id`. The AI/DL handover asks for “a stable `user_id` or persistent anonymous user key” but does not name one. The proposal is that both pipelines use `user_pseudo_id` as the anonymous key and add the logged-in `user_id` alongside it once it is captured. Both teams must use the same identity for a persona lookup to find the user DL-05 is scoring.
- **Shared source.** Both pipelines build from the same GA4 BigQuery export. The proposal is for DE to publish one normalised event table that both read, so that user counts and time windows reconcile.
- **Different time horizons, by design.** DL-05 looks at 7 days of history and scores products the user touched in the last 2 days. DA-01 aggregates over a longer reporting window and refreshes weekly. The persona is the slower-moving context, and DL-05's features are the in-the-moment signal. Neither replaces the other.

Proposed fallback order (replaces the v1.0 cold-start rules)

1. **The user has recent-interaction candidates:** DL-05's ranked Top-N. If it returns fewer than N products, top up the list from step 2.
2. **No candidates, and a DA-01 persona exists:** a persona-based Top-N built from the Section 4 candidate rules. Before product-level GA4 exists, these rules can draw on the product and pricing database and DA-03 True Value Promotion scores, which do not depend on GA4 product events.

3. **No DA-01 persona, but a T1 questionnaire answer exists:** a Top-N for the user's T1 Rule-Based Persona Framework persona (per DA-01-T2 and DA-01-T11 §9).
4. **Neither:** site-wide popularity or current specials. This is also the default for Low-Engagement Users (Section 4.3).

DL-05 decides the final fallback policy. The order above is DA/ML's proposed default.

Proposed refresh propagation

- On each new CSV publish, DL-05 reloads persona_label for the affected user_pseudo_ids before its next scheduled scoring run. The AI/DL handover already recommends batch scoring into a cached recommendation table, and persona reloads fit that schedule.
- cluster_id is NOT guaranteed stable across re-runs of the notebook (see FAQ Q3 in 6.5). Downstream systems must key on persona_label + assignment_date for any cross-run continuity claim, not on cluster_id alone.
- Latency target: personas from a Monday refresh should be reflected in DL-05 output by end-of-day Tuesday. Proposed; confirm in T3.

Open questions for the first T3 AI/DL checkpoint

- What counts as a "purchase" on DiscountMate? DL-05's training label is a purchase within 3 days, but DiscountMate is a comparison platform with no on-site checkout (DA-01-T3 §1). The label needs a DiscountMate proxy, such as Outbound_Merchant_Click, add-to-list or Add_To_Favourites. That choice shapes both teams' instrumentation ask (Section 8.4).
- Who owns the persona candidate rules in integration point 2? The proposal is that DA/ML drafts them from Section 4 and AI/DL implements them as a candidate generator.
- Where does the persona file live? Section 6.1 now proposes the same repository as DL-05.

6.3 Interface with DA-02 Insight Surfacing

DA-02 (Senushi's Power BI dashboard work) already consumes DA-01 clustering outputs for the persona visualisation view ([DA-02-T8] in the roadmap). This subsection confirms the interface for T3 use.

Proposed consumption

- DA-02 reads the same persona_assignments CSV described in 6.1, or (T3 preferred) pulls persona rollups from the DE silver layer if a persona dimension is materialised there.
- Persona rollups DA-02 has requested (or DA/ML anticipates it will need): per-persona user count and share, per-persona mean of each mm_ feature, assignment_date so the dashboard can show "as-of" context, and refresh_freshness (days since assignment_date).
- Persona colour and label conventions: DA-02 owns the visual layer; DA/ML asks only that the four labels in §4 are used verbatim and that Cluster 2 (Low-Engagement) is not colour-coded in a way that reads as "bad users" — this is an engagement descriptor, not a value judgement.

6.4 Interface with DA-03 Pricing Intelligence

DA-03-T12 (persona-aware TVP scoring variants, P2 in the T2 roadmap) is the downstream that needs persona labels most directly. Amoda leads DA-03; the interface below is DA/ML-internal but stated here for cross-workstream clarity.

Proposed consumption

- DA-03-T12 joins scored product rows against persona_assignments on user_pseudo_id to layer per-persona weighting on the core True Value Promotion score.
- Suggested per-persona weight starting points (DA-03 to tune): Deal & Comparison Shoppers — heavy weight on discount depth and unit-price fairness; Highly Engaged Planners — moderate weight, with an added preference for staples and list-relevant items; Product-Focused Explorers — light TVP weight, category/detail relevance dominates; Low-Engagement Users — TVP score suppressed in favour of popularity defaults (consistent with 6.2 cold-start behaviour).
- The 12 mm_ features on the CSV are available to DA-03 if a continuous weighting scheme (rather than four discrete per-persona weights) proves more effective.

6.5 Consumer FAQ

The following questions are anticipated from downstream engineers and analysts reading this spec cold. Answers reflect the DA/ML position as of T2 close; the AI/DL and downstream reviews in T3 may adjust them.

Q1. What is the refresh cadence, and how do I know when a new persona is available?

T2: manual re-run of the clustering notebook, on request to DA/ML. T3 target: weekly, aligned to the GA4 BigQuery export and DE silver-layer refresh (see 6.1). Consumers should key on assignment_date in the file and treat any row where (today – assignment_date) > 14 days as stale; behaviour on stale data is Q4 below.

Q2. What should I do for a user with no persona assignment (cold-start)?

Follow the fallback order in 6.2: DL-05 recent-interaction ranking; if there are no candidates, a Top-N for the user's DA-01 persona; if there is no DA-01 persona, the T1 Rule-Based Persona Framework (per Carlin's DA-01-T2 review); if neither exists, popularity or current specials. Low-Engagement Users go straight to popularity or specials. See §4.3 and 6.2.

Q3. Are cluster_ids stable across re-runs of the model?

No. K-means cluster_ids are arbitrary integer labels — the algorithm may return the same four groupings with cluster_id 0 and 2 swapped on a subsequent run. persona_label is the stable identity (DA/ML re-inspects the cluster profiles after each run and assigns labels per §4). Consumers should key on persona_label, not cluster_id, for any cross-run continuity claim. Where cluster_id is used, always pair it with assignment_date.

Q4. What do I do if the CSV is stale?

For DL-05: continue serving the last available persona for that user, and fall back to popularity defaults for any user whose last known persona is > 30 days old (proposed; AI/DL to confirm). For DA-02 dashboards: show a freshness indicator (days since assignment_date) so stakeholders can see the age of the data. For DA-03: TVP scores remain valid; the persona weighting layer degrades gracefully to the un-weighted score if persona is missing or stale.

Q5. How do I interpret the mm_ features?

The mm_ prefix means min-max scaled to [0, 1] across the 1,500-user modelling matrix (per DA-01-T3 §4.3). A value of 0.75 means the user sits at the 75th percentile of that feature across the modelling population — not that 75% of their sessions do the thing. Definitions of the underlying counts and ratios are in the DA-01-T7 data dictionary; feature construction is in DA-01-T3 §3 and in the clustering notebook.

Q6. What is the current sample the personas are trained on, and how confident should I be?

The personas ratified in §4 are trained on a 1,500-user synthetic dataset with market-research-grounded proportions (per Modelling Documentation and DA-01-T11 §8). The original real GA4 sample was too small (~28 users) for meaningful K-means or DBSCAN. The pipeline is validated on synthetic data; the personas

should be re-validated on real GA4 data in T3 as user volume grows. Downstream teams should treat the persona distribution as provisional and expect proportions to drift on real data.

Q7. Why isn't there a Premium Shopper persona? T1 had one.

No cluster in the K=4 solution is primarily characterised by premium-product preference, because the current feature set does not include category- or product-price signals (per DA-01-T11 §5.5). This is a Phase 1 feature-set limitation, not a data-quality issue. Phase 2 events (Product_Click, Retailer_Selected, etc.) will make premium detection possible in T3 or later.

Q8. Who do I talk to about changing this interface?

Section 6 is a proposal from the DA/ML side. For DL-05 interface changes: raise with AI/DL leads (Ethan Fu, Sunain Mushtaq, Lam Yu) and DA/ML (Robin Le, Shubhpreet Kaur). For DA-02 interface changes: Senushi (DA-02 sub-lead) and DA/ML. For DA-03 changes: Amoda Senarath (DA-03 lead). For the CSV schema, refresh cadence, or persona definitions themselves: DA/ML.

7. Assumptions & Limitations

These are the limits of a Phase 1 pipeline validated on synthetic data. Each one is stated with its effect on downstream teams and where it is addressed.

7.1 Data volume — synthetic modelling substrate

The model was fitted on a 1,500-user synthetic dataset that DE generated to market-research-grounded proportions. The real GA4 sample was 28 users (DA-01-T11 §8), below the ~50–100 minimum in DA-01-T3 §6. This validates that the pipeline runs end to end and produces separable, interpretable clusters. It does not show that the four labels are the segments DiscountMate’s real users will form: the cluster count, shapes and sizes could all change, and the silhouette score on real data is unknown. **Mitigation:** re-run the pipeline and re-tune K once real volume clears ~500 active users over 28 days (Section 8.5). Until then, treat the labels as directional segments.

7.2 Feature coverage — Phase 2 GA4 events not yet instrumented

The eight Phase 2 events (Product_Click, Add_To_Favourites, Price_Alert_Created, Retailer_Selected, Coupon_Click, Filter_Used, Sort_Used, Outbound_Merchant_Click) are not live. Their eight dependent features are held as nulls. This is the largest constraint on persona quality, and it is why the labels describe *what* users do rather than *why*. With the current features, Product-Focused Explorers cannot be split into Health Enthusiast and Convenience Seeker, Highly Engaged Planners cannot be confirmed as Family Planners, and no segment reproduces Premium Shopper (DA-01-T11 §5). The same gap blocks DL-05: it has no product-level GA4 events to build its features from (Section 6.2). **Mitigation:** one joint instrumentation ask serving both teams (Section 8.4). The DA-01 schema already reserves the columns.

7.3 Reporting window — static 28-day rolling, not real-time

Features are aggregated over a rolling window (DA-01-T3 §4.1), so a persona label is a snapshot. It updates on the refresh cadence, not on a user’s next session. That suits dashboards and cohort analysis. For recommendations, within-session responsiveness comes from DL-05’s own 7-day and 2-day features, with the persona supplying slower context (Section 6.2). **Mitigation:** scheduled refresh (weekly, nightly once on GCP) and drift monitoring (Section 9.2).

7.4 No purchase data linked to behavioural signals

All current features are browse-side. There is no linked signal showing whether a user bought anything, on DiscountMate or at a retailer. The personas therefore describe browsing patterns, not conversion. Any revenue-facing use (promotion targeting, TVP weighting, recommender evaluation) must assume that browsing predicts buying. The same gap means DL-05’s purchase label has no direct DiscountMate equivalent (Section 6.2, open questions). **Mitigation:** use Outbound_Merchant_Click as an interim proxy for purchase intent. True purchase-linked signals depend on receipt-based tracking, which is long-term work (Section 9.3).

7.5 Behavioural labels ≠ motivational personas

The four labels describe behaviour on the platform. They are not T1-style motivational personas. “Highly Engaged Planners” is compatible with Family Planner but does not mean Family Planner. Reading a behavioural label as a motivation overclaims what the model supports. **Mitigation:** downstream teams use the behavioural label as the identity and the T1 mapping in Section 4 as context. Motivational labels require Phase 2 events plus qualitative research (Sections 8.5 and 9).

8. Recommendations for Downstream Teams

One subsection per downstream consumer. Each recommendation is a concrete ask sized against the current state, not a wish list; every ask names the DA/ML side's obligation as well as the ask on the consumer.

Section 6 is the interface reference; this section is the practical use.

8.1 AI/DL — DL-05 Personalised Top-N Recommender

- **Agree the revised interface in Section 6.2 at the first T3 checkpoint.** Confirm the four integration points and the fallback order, standardise on `user_pseudo_id` as the shared join key, and settle the three open questions: the DiscountMate purchase proxy, ownership of the persona candidate rules, and the file location. DA/ML's obligation is to publish the persona file (Section 6.1) on the agreed cadence and draft the per-persona candidate rules from Section 4.
- **Fill the empty-candidate branch with the persona fallback first.** The draft serving code returns an empty list when a user has no recent interactions, and on the test period that covered most users (candidate-user coverage 13.4%). A persona-based Top-N, then a T1 RBPF Top-N, then popularity or specials (Section 6.2) addresses the AI/DL handover's Priority 7. It can run on the product and pricing database before product-level GA4 exists, so it is the earliest point at which the two features create value together.
- **Test personas as a candidate source, measured by coverage.** DL-05's ranking works well on products that reach the candidate set; the bottleneck is getting them there (coverage 11.7%). When broader candidate generation is built (Priority 6), add persona-based candidates as one source and report the coverage gain from that source on its own. Deal & Comparison Shoppers is the first segment to test: it is the largest (525 users, 35%), has the strongest T1 alignment, and has candidate rules (specials, high-TVP items) that can be built from pricing data now.
- **Report DL-05 metrics by persona.** Break NDCG@10, HitRate@10 and candidate coverage down by the four segments rather than reporting one global figure. The expected pattern is near-zero coverage for Low-Engagement Users, which is why popularity or specials is their default (Section 4.3), and the highest coverage for Highly Engaged Planners.
- **Redefine the training label for DiscountMate before retraining.** DL-05 predicts purchase within 3 days, but DiscountMate has no on-site checkout. Agree a proxy jointly (`Outbound_Merchant_Click`, `add-to-list` or `Add_To_Favourites`) so that the GA4 instrumentation ask in Section 8.4 captures it. The same decision determines DA-01's purchase-intent proxy (Section 7.4).
- **Add personas as a ranking feature only at the DiscountMate retrain, and only if the test shows a gain.** When the model is retrained on DiscountMate behaviour (Priority 5), compare the 55 features alone against the 55 plus cluster flags (and optionally the 12 `mm_` features) on NDCG@10 per segment. Keep persona inputs only if they add lift. Until then, the 55-feature contract stays unchanged.

8.2 DA-02 Insight Surfacing

- Dashboard the four segments as their own view. The Power BI segment overview and segment profiles pages already scoped in the Feature 2 wireframes are the right place; the persona rollups defined in Section 6.3 are the source. Include size (N and % of population), the top three defining `mm_` features per segment with direction (H/M/L), and the T1 alignment tag from Section 4.
- Track movement over time. Once a second refresh of the persona-assignment CSV lands, the delta between refreshes is the smallest unit of movement worth surfacing: how many users changed persona, and which transitions dominate (e.g., Low-Engagement Users → Deal & Comparison

Shoppers is a reactivation signal worth flagging to product). This is the operational form of the persona-lifecycle work in Section 9.2 and is the earliest place it becomes actionable.

- Use the persona rollups, not raw cluster IDs, in stakeholder-facing views. Section 6.3 defines the rollups (persona label, per-persona count and share, per-persona feature means, assignment date). Cluster IDs are stable across refreshes only while the model is stable; if K-means is re-fit on real T3 data and the label ordering changes, dashboards keyed on cluster ID silently break, whereas dashboards keyed on rollup label degrade visibly.

8.3 DA-03 Pricing Intelligence

- Use persona labels for the persona-aware TVP scoring variant ([DA-03-T12]). This is the P2 extension already scoped in the roadmap; Deal & Comparison Shoppers is the segment where a persona-weighted TVP score has the highest expected value, because the segment's Compare/Specials signature is directly relevant to whether a promotion registers as good value.
- Co-develop a draft persona weighting table in T3. A first-pass weighting — Deal & Comparison Shoppers weights discount % and cross-retailer savings; Highly Engaged Planners weights staples and repeat-buy value; Product-Focused Explorers weights product quality/attribute signals over discount %; Low-Engagement Users uses the unweighted default — is a good starting point for A/B evaluation. DA/ML will hold the persona-assignment feed; DA-03 owns the weighting logic; both sides align on the schema before implementation to avoid a mid-T3 re-fit.
- Do not persona-condition the TVP core score itself. The persona-aware variant is a layer on top of the base True Value Promotion V2 output, not a replacement for it. This preserves TVP V2 as the interpretable, reproducible baseline downstream of any persona churn.

8.4 App Dev / Data Engineering — Phase 2 GA4 instrumentation

- Make it one joint instrumentation ask for DA-01 and DL-05. The AI/DL recommender handover names product-level GA4 tracking as its Priority 1 blocker. Every product interaction needs a stable user key, timestamp, action, item ID mapped to the canonical DiscountMate product ID, session ID, category, brand and price. The events below, carrying those item parameters, serve both teams. The purchase proxy agreed in Section 8.1 decides which event DL-05 trains on.
- Instrument the eight missing Phase 2 events. In priority order: Product_Click (unlocks category affinity, the single highest-leverage feature for distinguishing Health Enthusiast, Convenience Seeker, and Premium Shopper); Retailer_Selected (unlocks retailer diversity, directly relevant to Budget-Conscious Shopper alignment); Filter_Used and Sort_Used (unlock discovery-behaviour features and are cheap to instrument); Add_To_Favourites and Price_Alert_Created (unlock preference and price-sensitivity signals); Coupon_Click and Outbound_Merchant_Click (unlock promotion engagement and purchase-intent proxies).
- This is the single highest-leverage unlock for persona quality in T3. Volume alone (7.1) will sharpen the four current labels; Phase 2 instrumentation is what lets the model resolve motivational-style personas the T1 framework anticipates but that the current feature set cannot support (7.2, 7.5). Both are needed; instrumentation is the one that has been named in every DA-01 spec since DA-01-T3 and has not moved.
- Data Engineering: preserve the current DA-01 schema contract. The clustering-ready dataset schema (DA-01-T3 §5.1) already reserves the null columns for Phase 2 features; extraction pipelines should populate them as-and-when events land rather than treating instrumentation as a schema change.

8.5 Incoming T3 Feature 1 lead

- Re-run the pipeline against real GA4 data once volume clears ~500 active users over a 28-day window. Below that, the K-means fit is dominated by whichever handful of users happen to be in-window; above it, silhouette and elbow become interpretable. This is the criterion for promoting the four labels from "validated on synthetic data" to "validated on real behaviour".
- Re-tune K rather than assuming K=4 holds. The elbow+silhouette procedure in Section 5.3 is the right procedure; run it on the real data and let the result stand. If K remains at 4, the four current labels are the working hypothesis for interpretation; if K shifts to 3 or 5, the interpretation step (DA-01-T11 procedure) applies again to the new solution.
- Validate the four labels against real behaviour. For each of the current four labels, the DA-01-T11 procedure — feature-signature summary, T1 mapping with alignment tag, note on what current features can and cannot claim — is the template. Report where the labels hold, where they split, and where they cross. All three outcomes are informative and none is a failure state (this is inherited from DA-01-T3 §7 and worth restating).
- Preserve the honest framing. "Pipeline validated on synthetic data conforming to spec" is not the same claim as "discovered real user personas" and should not drift into it during T3 write-ups. This has been a recurring correction throughout T2 and belongs in the T3 lead's brief on day one.

9. Roadmap Forward

Three horizons, in ascending order of ambition and prerequisite. Near term is the T3 backlog that immediately follows this handover; medium term is the work that becomes possible once T3 has real data and Phase 2 events; long term is the vision inherited from the T1 handover (Section 15.1 strategic context) and shared with Data Engineering and AI/DL.

9.1 Near term (T3)

- Real-data revalidation. Re-run the pipeline against real GA4 data at the volume threshold in Section 8.5; re-tune K; re-check the four labels. Owner: incoming T3 Feature 1 lead. Prerequisite: ~500 active users over 28 days; likely reachable within T3 given the W6 onboarding push planned in the T2 roadmap.
- Phase 2 features. Once the eight events land (Section 8.4), extend the feature space from 12 to ~20 mm_ features, re-fit the model, and update the persona cards. The clustering-ready schema already reserves the columns; the DA-01-T4 extraction and DA-01-T6 normalisation steps handle new columns without re-architecting.
- DL-05 integration. Agree the revised interface with AI/DL (Section 8.1). Ship the persona fallback into DL-05's empty-candidate branch first, since it needs no product-level GA4. Then add persona candidates as a source once product events flow, with Deal & Comparison Shoppers as the first segment. Report DL-05 metrics by persona from the first evaluation run.
- Feedback loop from recommender clicks back into features. Once DL-05 is producing recommendations and users are clicking them, click-through on persona-sourced and DL-05-ranked recommendations becomes an additional behavioural signal that can feed the next T3 refit — a Product_Click-adjacent event under DA/ML ownership rather than pure GA4 instrumentation.

9.2 Medium term

- Temporal cluster drift monitoring. As refreshes accumulate, track how per-cluster centroids and sizes move over time. A slow drift in the Deal & Comparison Shoppers centroid, for example, is an early signal that promotion behaviour on the platform is changing; a fast drift is a data-quality alarm. This is the operational form of the "static 28-day window" mitigation in Section 7.3.
- Persona lifecycle. Rather than modelling each refresh as an independent snapshot, model how individual users move between segments over time: Low-Engagement Users → Product-Focused Explorers is an onboarding-success signal; Deal & Comparison Shoppers → Low-Engagement Users is a churn precursor. Section 8.2 flags this as the earliest useful surface in Feature 2.
- Category-affinity vectors. Once Product_Click is instrumented and volume allows, the category_affinity_vector feature in DA-01-T3 §3.5 becomes buildable. This is the single feature that most directly closes the Health Enthusiast / Convenience Seeker / Premium Shopper resolution gap identified in Section 7.5.

9.3 Long term

The long-term vision, carried from the T1 handover, is a hybrid persona system that combines rule-based signals (transparent, cold-start-friendly, inherited from the T1 RBPF), behavioural signals (this workstream), and purchase-driven signals (dependent on receipt-based product tracking flagged as future work in the T1 handover). Dynamic re-scoring — a user's persona confidence updates as their behaviour and purchases evolve, rather than being fixed at assignment — is the operational form of that vision and the natural evolution of the static-window refresh model described in Section 7.3.

None of this is scoped for T3. It is documented here so that the T3 lead's architectural choices — how the persona-assignment artefact is structured, how confidence is represented, how the interface with AI/DL is drawn — leave room for a hybrid system to grow into rather than a purely behavioural one to be re-built around.

10. Appendices

Appendix A — Data dictionary reference

The authoritative definitions of the behavioural features are in **DA-01-T7 Behavioural Data Dictionary** (Noel Mathew Shillow). It records each feature's definition, GA4 source event, data type and unit, derivation and interpretation direction. It also records the dataset grain (one row per included user), the GA4 source tables (events_YYYYMMDD and pseudonymous_users_YYYYMMDD from the BigQuery export) and the metadata columns (user_pseudo_id, reporting_window_start, reporting_window_end, is_included_in_run).

Read DA-01-T7 alongside Section 5.1 of this document. DA-01-T7 defines 13 features. The T2 clustering run used 12 of them: avg_percent_scrolled is excluded for the reason given in Section 5.1. The mm_ columns in the persona-assignment file (Section 6.1) are the min-max-scaled versions of those 12 features. Feature construction logic is in DA-01-T3 §3 and the clustering notebook.

- File: DA-01-T7_Behavioural_Data_Dictionary.docx (Feature 1 handover folder; see Appendix B)

Appendix B — Notebook & artefact index

Artefact	File / location	Notes
Feature 1 handover folder	Feat-1-GA4-segmentation (DA/ML shared working folder)	Holds the local copies of the files listed below
Clustering notebook (DA-01-T9/T10)	Handover folder: DA-01-T9-Clustering_Model.ipynb. Proposed repository path: notebooks/DiscountMate_Clustering_Model.ipynb	Same notebook under two names. Export cell for the persona-assignment file still to be added (Section 5.3)
Modelling Documentation (DA-01-T10)	DA-01-T10-Modelling-Documentation.docx (also circulated as Modelling_Documentation.pdf)	K selection, cluster profiles, PCA, DBSCAN result
Cluster interpretation (DA-01-T11)	DA-01-T11_Interpret_Behavioural_Clusters_vs_T1_Personas.pdf	Source for the Section 4 persona cards
Feature specification (DA-01-T3)	DA-01-T3_Behavioural_Feature_Specification_v1.0.docx	Features, preprocessing, clustering requirements
Data dictionary (DA-01-T7)	DA-01-T7_Behavioural_Data_Dictionary.docx	See Appendix A
T1 framework review (DA-01-T2)	DA-01-T2_Technical_Review_Rule-Based_Persona_Framework.docx	Basis for Section 3.2
GA4 event audit (DA-01-T1)	DA-01-T1_GA4_Event_Audit_Report.pdf	Source of the Phase 2 event list
Clustering-ready dataset sample (DA-01-T6)	DA-01-T6-Normalized-clustering-ready-synthetic-dataset.csv	Normalised synthetic sample
Persona archetype definitions	DA-01_Persona_Archetype_Definitions_1.docx	Archetype targets given to DE for synthetic data
Synthetic behavioural dataset	synthetic_behavioural_data/ (synthetic_users_20260831.csv, generate_discountmate_synthetic_users.py, README.md)	1,500 users, fixed random seed, method note in README
Persona visualisation (DA-	DA-02-T8-Persona-Visualisation-Prototype-	Power BI prototype of the DA-01

Artefact	File / location	Notes
02-T8)	Batnasan 1.pbix	clusters
T2 2026 Roadmap	T2 2026 Roadmap.docx	Source for Appendix C
Persona-assignment output (proposed)	GitHub repository DataBytes-Organisation/DiscountMate_new (DA/ML path to agree with AI/DL): persona_assignments_YYYYMMDD.csv and personas_latest.csv	Proposed in Section 6.1; not yet published
DL-05 recommender implementation (AI/DL)	DataBytes-Organisation/DiscountMate_new, ML/Recommendation_system/Recommendation-2026-t2/ (merged via PR #319); model bundle in models/	Main notebook: DiscountMate_additional_features_model.ipynb

Appendix C — Task ledger, DA-01-T1 → T12

Task titles, weeks and priorities are taken from the T2 2026 Roadmap (Section 5.4, Feature 1). Owners are as recorded in Section 5.5 and on each artefact. SP means story points.

Task ID	Title	Week	Priority / SP	Status	Owner	Artefact reference
DA-01-T1	Audit existing GA4 events and identify missing behavioural signals	W4	P0 / 2	Complete	Noel Mathew Shillow	DA-01-T1 GA4 Event Audit Report
DA-01-T2	Review existing rule-based persona framework and Adam Creek's T1 exploratory doc	W4	P0 / 2	Complete	Carlin Monsy	DA-01-T2 Technical Review
DA-01-T3	Define behavioural feature list and clustering requirements	W4–5	P0 / 3	Complete	Robin Le	DA-01-T3 Specification v1.0
DA-01-T4	Extract GA4 behavioural data and prepare initial dataset	W5	P0 / 3	Complete (real GA4 sample of 28 users; see Section 7.1)	Carlin Monsy	Initial behavioural dataset
DA-01-T5	Clean behavioural dataset (invalid records, duplicates, noise)	W5–6	P0 / 3	Complete	Tanvi	Cleaned dataset
DA-01-T6	Prepare and normalise the clustering-ready dataset	W6	P0 / 3	Complete	Shubhpreet Kaur	DA-01-T6 clustering-ready dataset
DA-01-T7	Document behavioural signals; create data dictionary	W5–6	P0 / 2	Complete	Noel Mathew Shillow	DA-01-T7 Data Dictionary
DA-01-T8	Validate the behavioural dataset before clustering	W6	P0 / 2	Complete	Feature 1 team	Validation checks (Section 5.5)
DA-01-T9	Develop the segmentation model (K-means and/or DBSCAN)	W7–8	P1 / 8	Complete (on 1,500-user synthetic dataset)	Shubhpreet Kaur	Clustering notebook
DA-01-T10	Evaluate cluster quality	W8	P1 / 3	Complete (K=4,	Shubhpreet	Modelling

Task ID	Title	Week	Priority / SP	Status	Owner	Artefact reference
	and select results			silhouette $\approx$ 0.484)	Kaur	Documentation
DA-01-T11	Interpret clusters and compare against T1 personas	W8–9	P1 / 5	Complete; labels ratified by Feature 1 team	Robin Le	DA-01-T11 report
DA-01-T12	Finalise persona definitions; document methodology and results	W9–10	P1 / 3	Complete (this document, v1.0)	Robin Le	This document

Roadmap checkpoint not met in T2. The roadmap set a cross-team touchpoint for the persona-to-recommender interface to be confirmed with AI/DL by W9. That confirmation did not happen within T2. v1.1 revises the proposal against the AI/DL recommender handover, but Section 6 remains a proposal, and agreeing it is the first T3 action (Section 8.1).

Appendix D — Glossary

Term	Definition
RBPF	Rule-Based Persona Framework. The T1 approach that assigns users to one of five predefined personas from an onboarding questionnaire using deterministic rules.
GA4	Google Analytics 4. The web analytics platform that records DiscountMate user events (page views, sessions, engagement) and exports them to BigQuery.
K-means	A clustering algorithm that splits users into K groups by assigning each user to the nearest group centre. It is the operational model for Feature 1, at K=4.
DBSCAN	Density-Based Spatial Clustering of Applications with Noise. A clustering algorithm that finds dense regions and labels sparse points as noise. It was tested as an alternative and not adopted (Section 5.3).
Silhouette score	A measure of how well separated clusters are, from –1 to 1; higher is better. The Feature 1 solution scores $\approx$ 0.484 (moderate).
PCA	Principal Component Analysis. A technique that compresses many features into a few summary dimensions. It was used here only to visualise the clusters in two dimensions.
mm_ prefix	Marks a feature min-max scaled to [0, 1] across the modelling population. For example, 0.75 means the 75th percentile of that feature, not 75% of activity.
Cold-start	The problem of personalising for a user with no behavioural history. In Feature 1 it is handled by falling back to the RBPF, then to popularity-based defaults.
Phase 1 features	The 12 behavioural features buildable from GA4 events already live: engagement, page-type affinity, browsing breadth, recency and tenure. Used in the T2 model.
Phase 2 features	Features that depend on eight GA4 events not yet instrumented (Product_Click, Add_To_Favourites, Price_Alert_Created, Retailer_Selected, Coupon_Click, Filter_Used, Sort_Used, Outbound_Merchant_Click). Reserved in the schema as nulls.
DL-05	The AI/DL team's Personalised Top-N Recommender, the primary consumer of the personas.
TVP	True Value Promotion. DA-03's score for how genuinely good a promotion is. DA-03-T12 is the persona-aware variant.
BigQuery	Google Cloud's data warehouse, which receives the daily GA4 event export.
GCP	Google Cloud Platform, where scheduled refresh jobs are proposed to run (Section 6.1).
UX	User experience.

Term	Definition
UTC / RFC 4180 / BOM	File conventions in Section 6.1: Coordinated Universal Time; the standard CSV format specification; byte-order mark (a hidden marker at the start of a text file, excluded here).
REES46	The public e-commerce behaviour dataset (October 2019) on which DL-05 was trained.
Candidate generation / candidate coverage	The step that picks which products a recommender will score for a user, and the share of relevant items or users that step reaches. DL-05's coverage is 11.7% of pairs.
NDCG@10 / HitRate@10	Ranking-quality metrics for a Top-10 list: NDCG rewards putting relevant items near the top; HitRate is the share of users with at least one relevant item in their Top-10.
PBIX	The Power BI Desktop file format, used for the DA-02 dashboards.
SP / Wn	Story points (relative effort estimate) / week <i>n</i> of the trimester.

Appendix E — Change log

Version	Date	Author	Change
0.1	17 Sep 2026	Robin Le	Wave 1 draft: Section 4 (Final Persona Definitions) and Section 6 (Downstream Interface Specification).
0.2	22 Sep 2026	Robin Le	Wave 2 draft: Sections 5, 7, 8 and 9.
1.0	24 Sep 2026	Robin Le	Initial release. Added Sections 0–3 and 10; assembled Sections 0–10. Consolidation corrections: removed wave scope notes; aligned cross-references (Sections 5.1, 5.2, 5.3, 8.2, 8.4); corrected the source of the avg_percent_scrolled exclusion note; expanded acronyms on first use. Approved by the Feature 1 team.
1.1	24 Sep 2026	Robin Le	Revised against the AI/DL Personalised Recommendation Engine T2 handover: rewrote Section 6.2 (integration points, join key, fallback order, open questions) and Section 8.1. Updated the persona recommender implications in Section 4, file location in 6.1, FAQ Q2, and Sections 8.4 and 9.1 (joint GA4 product-level instrumentation ask). Aligned Sections 1, 2.3, 2.4 and 5.5. Condensed Section 7. Added the AI/DL handover to the related artefacts, Appendix B and the glossary.